BlindSpot Robustness Audit: exp_1790977223_6c3ec3

Experiment Run: **exp_1790977223_6c3ec3** | Multi-Model Diagnostic Certification Report

Created: 1790977224.404111	**Evaluated Models:** 5	**Integrity Status:** PASS
Total Diagnosed Failures: 4	**Probes per Model:** 7	**Audit Framework:** Deterministic Behavioral + Attribution XAI

1. Model Architecture Baselines

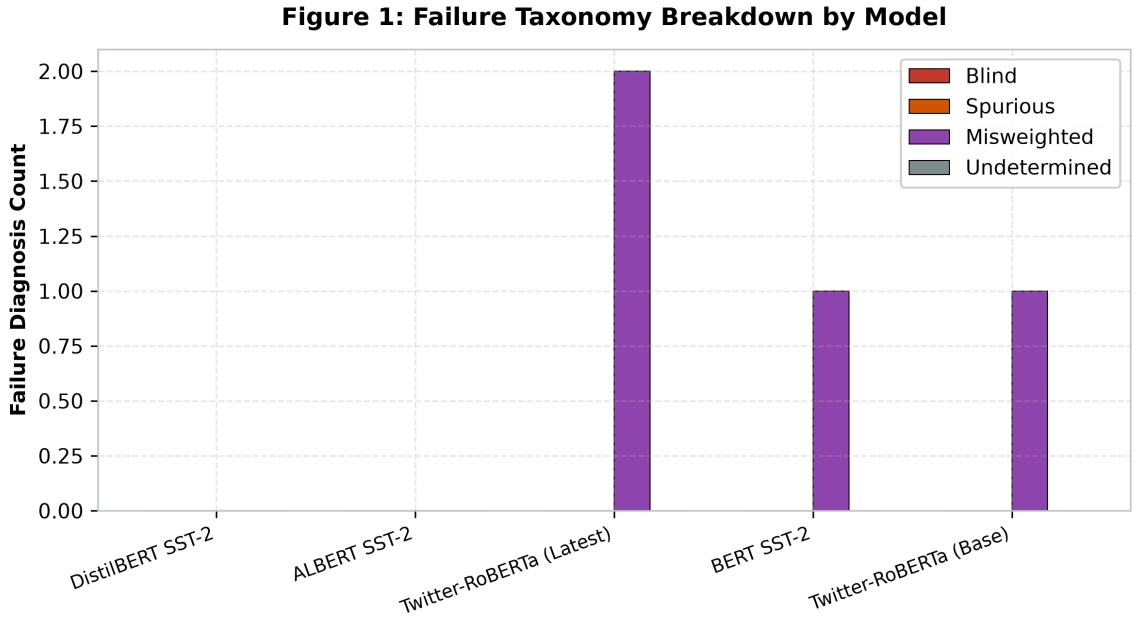
Model	Baseline Class	Confidence
distilbert-base-uncased-finetuned-sst-2-english	**POSITIVE**	0.9915 (99.2%)
albert-base-v2-SST-2	**POSITIVE**	0.7930 (79.3%)
twitter-roberta-base-sentiment-latest	**POSITIVE**	0.7435 (74.3%)
bert-base-uncased-SST-2	**POSITIVE**	0.8466 (84.7%)
twitter-roberta-base-sentiment	**POSITIVE**	0.6672 (66.7%)

2. Primary Research Metrics & Probabilistic Calibration

Model Architecture	Probes	Observed Flips	Expected Flips	Preserve %	Consistency %	ECE	Mean Δ Conf
distilbert-base-uncased-finetuned-sst-2-english	7	2 (28.6%)	28.6%	100.0%	**100.0%**	0.0371	-2.9 pp
albert-base-v2-SST-2	7	2 (28.6%)	28.6%	100.0%	**100.0%**	0.2414	-3.4 pp
twitter-roberta-base-sentiment-latest	7	1 (14.3%)	28.6%	100.0%	**71.4%**	0.2284	-9.4 pp
bert-base-uncased-SST-2	7	3 (42.9%)	28.6%	80.0%	**85.7%**	0.0789	+6.5 pp
twitter-roberta-base-sentiment	7	1 (14.3%)	28.6%	100.0%	**85.7%**	0.2233	-3.3 pp

3. Behavioral Failure Taxonomy & Root Cause Distribution

Figure 1: Failure Taxonomy Breakdown by Model



What this figure is: Grouped bar chart displaying the count of diagnosed behavioral failure modes (Blind, Spurious, Misweighted, and Undetermined) across evaluated models.

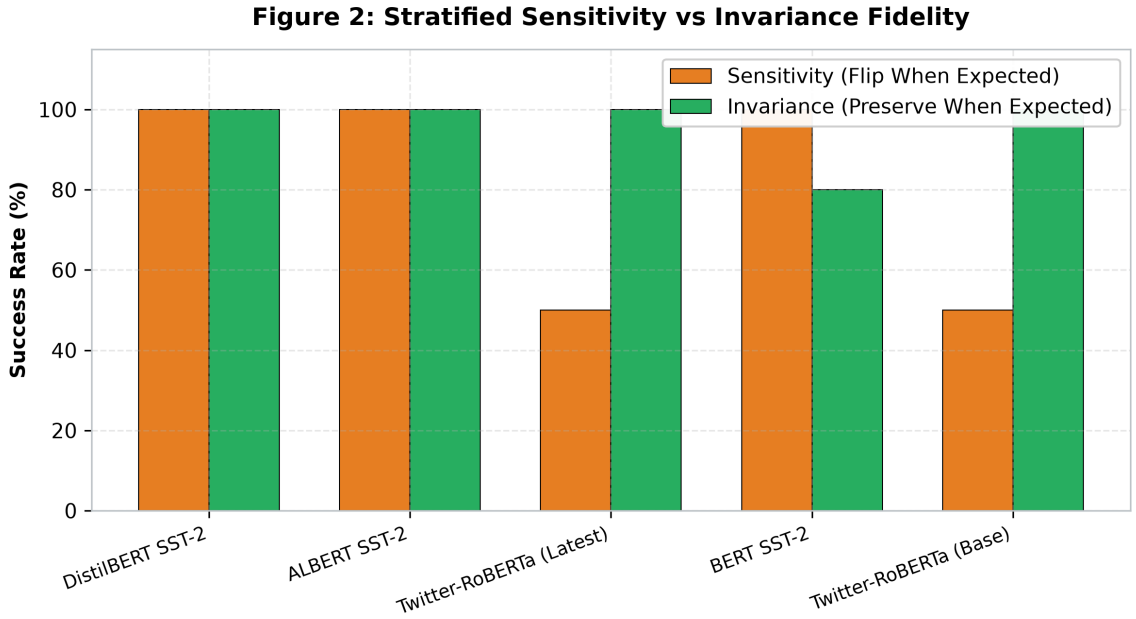
Empirical Observation: Across 5 evaluated models, a total of 4 behavioral failure(s) were diagnosed: 0 Blind, 0 Spurious, and 4 Misweighted. Failures were observed in: bert-base-uncased-SST-2, twitter-roberta-base-sentiment, twitter-roberta-base-sentiment-latest.

Diagnostic Significance: Distinguishes between attention blindness (ignoring operators) and feature misweighting (opposing tokens overpowering operators), guiding targeted architectural and data remediation.

Model Architecture	Blind	Spurious	Misweighted	Undetermined	Compliant (None)
distilbert-base-uncased-finetuned-sst-2-english	0	0	0	0	**7**
albert-base-v2-SST-2	0	0	0	0	**7**
twitter-roberta-base-sentiment-latest	0	0	2	0	**5**
bert-base-uncased-SST-2	0	0	1	0	**6**
twitter-roberta-base-sentiment	0	0	1	0	**6**

4. Sensitivity vs Invariance Fidelity & Confidence Dynamics

Figure 2: Stratified Sensitivity vs Invariance Fidelity

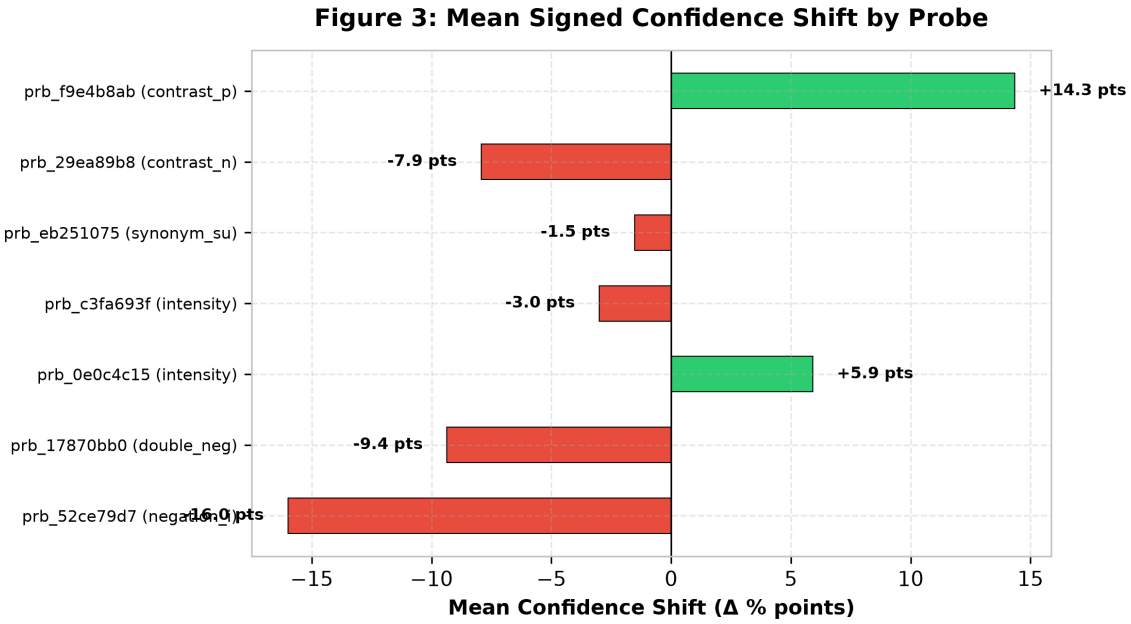


What this figure is: Dual-metric grouped bar chart contrasting Sensitivity (success rate in flipping on meaning-altering probes, in orange) against Invariance (success rate in preserving on meaning-preserving probes, in green).

Empirical Observation: Cohort sensitivity (flip compliance on meaning-altering probes) averaged 80.0%, while invariance fidelity (preservation rate on meaning-preserving probes) averaged 96.0%.

Diagnostic Significance: Separates responsive sensitivity to genuine semantic reversals from passive inertia under superficial or invariant linguistic modifications.

Figure 3: Mean Signed Confidence Shift by Probe



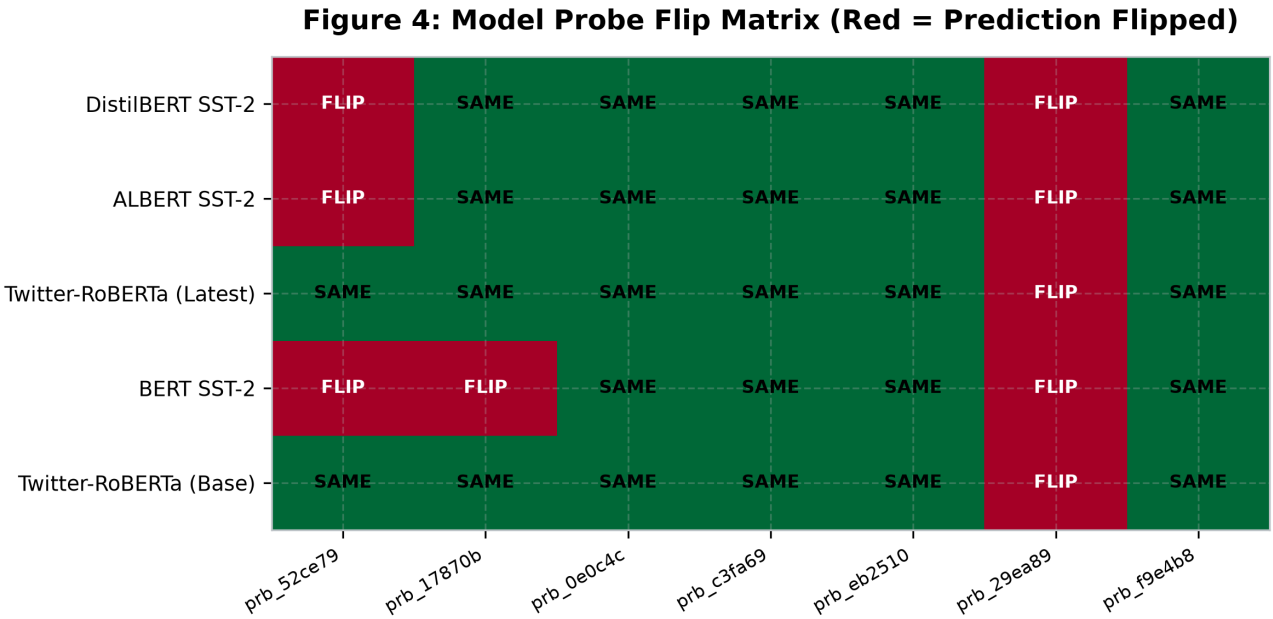
What this figure is: Horizontal divergence bar chart measuring the mean signed change in prediction confidence in percentage points (Δ Conf = p_pert - p_orig) for evaluated probes.

Empirical Observation: Mean signed confidence shift across evaluated model architectures spans from -9.4 pp to +6.5 pp (aggregate mean: -2.5 pp). Directional shifts reflect probe category impacts on model certainty.

Diagnostic Significance: Exposes sub-threshold prediction degradation where confidence significantly erodes under invariant edits or fails to attenuate under mitigating operators.

5. Cross-Model Concordance & Architectural Stability

Figure 4: Model Probe Flip Matrix

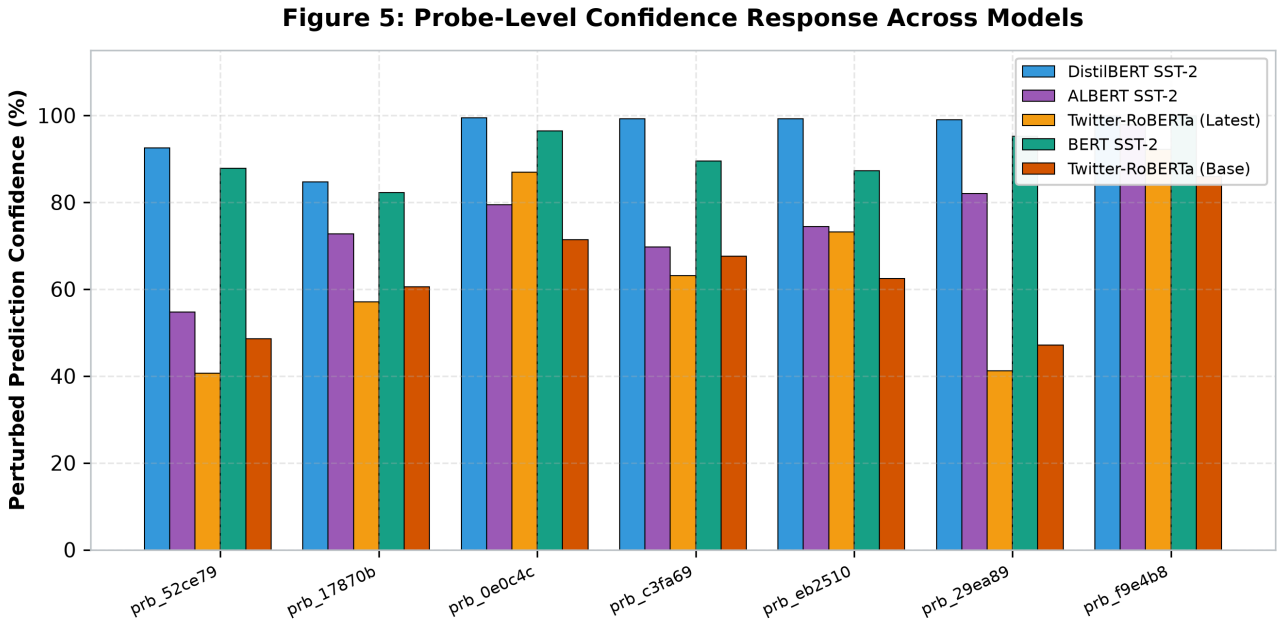


What this figure is: 2D heatmap matrix (Probes on horizontal axis, Models on vertical axis) displaying discrete prediction stability: Red indicates a prediction label flip (FLIP); Green indicates prediction stability (SAME).

Empirical Observation: Cross-model stability matrix mapping prediction flips across all evaluated probes. Highlights shared flips versus isolated architecture-specific label transitions.

Diagnostic Significance: Visualizes cross-architecture concordance, isolating systemic failure modes shared across model families from isolated single-model anomalies.

Figure 5: Probe-Level Confidence Response Across Models



What this figure is: Grouped bar chart displaying the perturbed prediction confidence percentage across all models for evaluated diagnostic probes.

Empirical Observation: Evaluates 7 diagnostic probe instances side-by-side across architectures, revealing variations in perturbed confidence levels and identifying linguistic triggers that cause divergent responses.

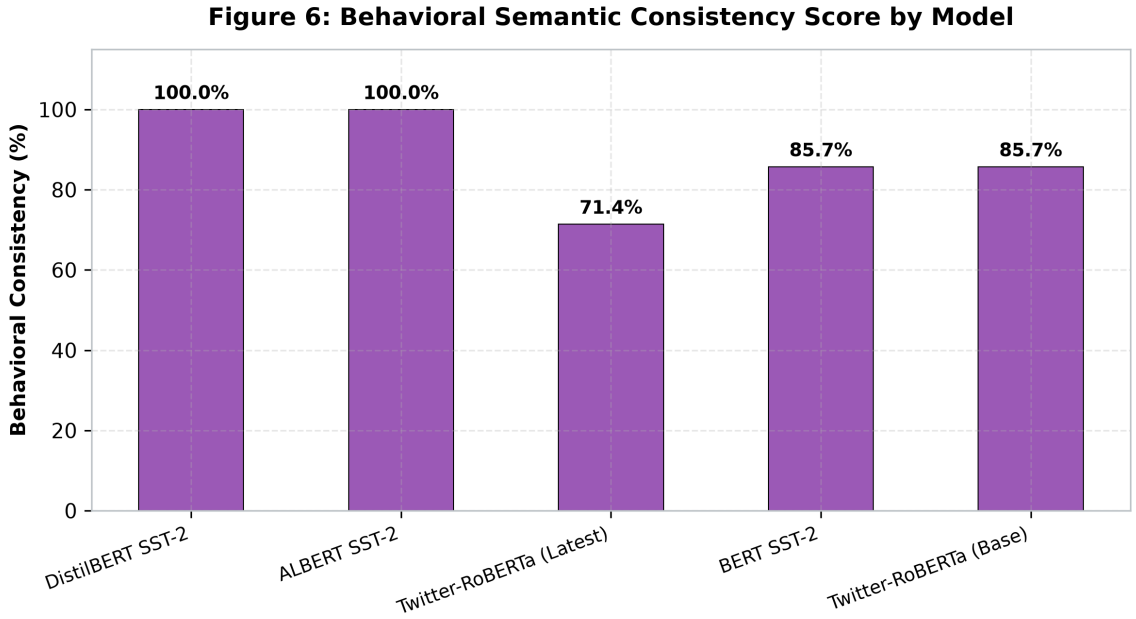
Diagnostic Significance: Provides probe-by-probe comparative resolution, identifying specific linguistic triggers that elicit architecture-specific confidence collapse or stability.

Pairwise Prediction Agreement Matrix (%)

Model	Twitter-RoBERT	Twitter-RoBERT	DistilBERT SST	ALBERT SST-2	BERT SST-2
Twitter-RoBERTa (Bas	100.0%	100.0%	85.7%	85.7%	71.4%
Twitter-RoBERTa (Lat	100.0%	100.0%	85.7%	85.7%	71.4%
DistilBERT SST-2	85.7%	85.7%	100.0%	100.0%	85.7%
ALBERT SST-2	85.7%	85.7%	100.0%	100.0%	85.7%
BERT SST-2	71.4%	71.4%	85.7%	85.7%	100.0%

6. Behavioral Consistency & Multi-Metric Profiles

Figure 6: Behavioral Semantic Consistency Score by Model

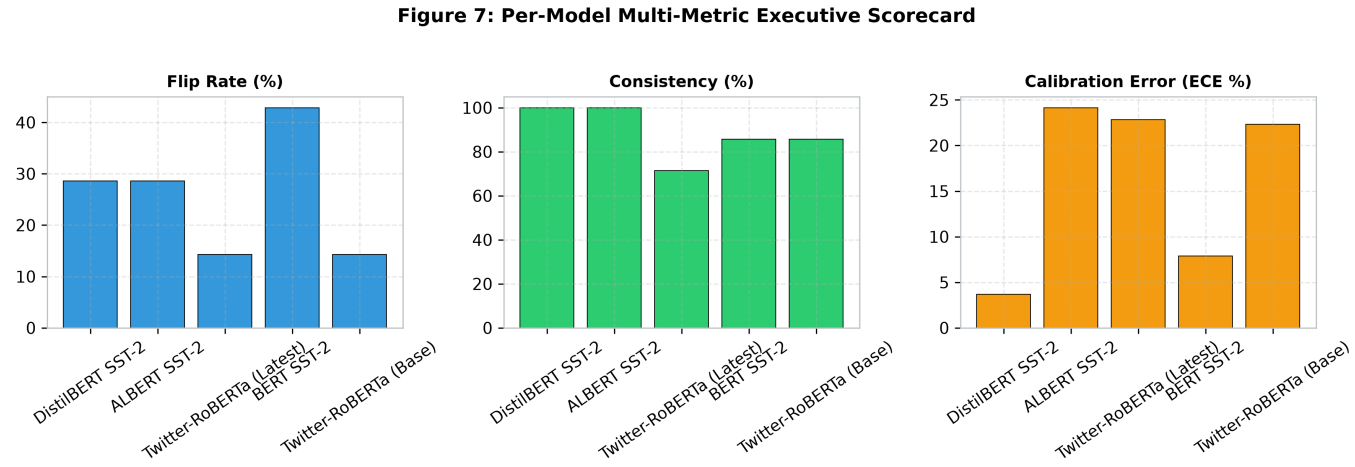


What this figure is: Bar chart illustrating the overall Behavioral Consistency percentage achieved by each evaluated model, measuring strict compliance with expected semantic effects.

Empirical Observation: Behavioral consistency across models ranges from 71.4% to 100.0% (cohort mean: 88.6%). Consistency measures empirical alignment between observed transitions and expected semantic effects.

Diagnostic Significance: Quantifies the proportion of probes where observed transitions obeyed the semantic contract (flipping when polarity is inverted, preserving when invariant).

Figure 7: Per-Model Multi-Metric Executive Scorecard



What this figure is: Three-panel comparative scorecard presenting Prediction Flip Rate (%), Behavioral Consistency (%), and Expected Calibration Error (ECE %) across models.

Empirical Observation: Executive multi-metric panel comparing flip rate, consistency score, and Expected Calibration Error (ECE ranges from 0.0371 to 0.2414).

Diagnostic Significance: Provides a tripartite summary of operational robustness, semantic compliance, and probabilistic calibration across all evaluated architectures.

7. Itemized Behavioral Failure Diagnoses & Explanations

Failure #1: [MISWEIGHTED] — Model twitter-roberta-base-sentiment-latest

• **Probe ID:** `prb_52ce79d7f530` (N/A)

• **Observed Transition:** `POSITIVE (0.74)` → `NEUTRAL (0.41)`

• **Diagnostic Reason:** Model recognized linguistic operator 'not' (LIME: -0.2805, SHAP: -0.4140), but opposing sentiment features ('good' (+0.2484)) outweighed the operator, preventing expected polarity reversal. Counterfactual operator ablation restored the baseline target probability direction, supporting operator sensitivity.

• **XAI Attribution Evidence:** Under negation_insertion, model transition 'POSITIVE -> NEUTRAL' was diagnosed as Misweighted. Model recognized linguistic operator 'not' (LIME: -0.2805, SHAP: -0.4140), but opposing sentiment features ('good' (+0.2484)) outweighed the operator, preventing expected polarity reversal. Counterfactual operator ablation restored the baseline target probability direction, supporting operator sensitivity.

• **Remediation Recommendation:** Augment training data and regularize behavior against negation_insertion perturbations to fix Misweighted failures.

Failure #2: [MISWEIGHTED] — Model twitter-roberta-base-sentiment-latest

• **Probe ID:** `prb_17870bb01cb0` (N/A)

• **Observed Transition:** `POSITIVE (0.74)` → `POSITIVE (0.57)`

• **Diagnostic Reason:** Polarity preserved, but confidence collapsed by 17.22 percentage points under meaning-preserving perturbation.

• **XAI Attribution Evidence:** Under double_negation, model transition 'POSITIVE -> POSITIVE' was diagnosed as Misweighted. Polarity preserved, but confidence collapsed by 17.22 percentage points under meaning-preserving perturbation.

• **Remediation Recommendation:** Augment training data and regularize behavior against double_negation perturbations to fix Misweighted failures.

Failure #3: [MISWEIGHTED] — Model bert-base-uncased-SST-2

• **Probe ID:** `prb_17870bb01cb0` (N/A)

• **Observed Transition:** `POSITIVE (0.85)` → `NEGATIVE (0.82)`

• **Diagnostic Reason:** Model unexpectedly flipped prediction under meaning-preserving perturbation due to disproportionate feature weight shift on semantic tokens.

• **XAI Attribution Evidence:** Under double_negation, model transition 'POSITIVE -> NEGATIVE' was diagnosed as Misweighted. Model unexpectedly flipped prediction under meaning-preserving perturbation due to disproportionate feature weight shift on semantic tokens.

• **Remediation Recommendation:** Augment training data and regularize behavior against double_negation perturbations to fix Undetermined failures.

Failure #4: [MISWEIGHTED] — Model twitter-roberta-base-sentiment

• **Probe ID:** `prb_52ce79d7f530` (N/A)

• **Observed Transition:** `POSITIVE (0.67)` → `NEUTRAL (0.49)`

• **Diagnostic Reason:** Model recognized linguistic operator 'not' (LIME: -0.2929, SHAP: -0.3933), but opposing sentiment features ('good' (+0.2199)) outweighed the operator, preventing expected polarity reversal. Counterfactual operator ablation restored the baseline target probability direction, supporting operator sensitivity.

• **XAI Attribution Evidence:** Under negation_insertion, model transition 'POSITIVE -> NEUTRAL' was diagnosed as Misweighted. Model recognized linguistic operator 'not' (LIME: -0.2929, SHAP: -0.3933), but opposing sentiment features ('good' (+0.2199)) outweighed the operator, preventing expected polarity reversal. Counterfactual operator ablation restored the baseline target probability direction, supporting operator sensitivity.

• **Remediation Recommendation:** Augment training data and regularize behavior against negation_insertion perturbations to fix Misweighted failures.

8. Granular Probe-Level Empirical Evidence Matrix

Model	Probe ID	Category	Observed Transition	Δ Conf	Flip	Outcome	Diagnosis
distilbert-base-un	prb_52ce	NEGATION_INSER	POSITIVE → NEGATIVE	-6.6	YES	EXPECTED_FLIP	None
distilbert-base-un	prb_1787	DOUBLE_NEGATIO	POSITIVE → POSITIVE	-14.4	NO	EXPECTED_PRESERVE	None
distilbert-base-un	prb_0e0c	INTENSITY	POSITIVE → POSITIVE	+0.3	NO	EXPECTED_PRESERVE	None
distilbert-base-un	prb_c3fa	INTENSITY	POSITIVE → POSITIVE	+0.0	NO	EXPECTED_PRESERVE	None
distilbert-base-un	prb_eb25	SYNONYM_SUBSTI	POSITIVE → POSITIVE	+0.1	NO	EXPECTED_PRESERVE	None
distilbert-base-un	prb_29ea	CONTRAST_NEGAT	POSITIVE → NEGATIVE	-0.2	YES	EXPECTED_FLIP	None
distilbert-base-un	prb_f9e4	CONTRAST_POSIT	POSITIVE → POSITIVE	+0.8	NO	EXPECTED_PRESERVE	None
albert-base-v2-SST	prb_52ce	NEGATION_INSER	POSITIVE → NEGATIVE	-24.6	YES	EXPECTED_FLIP	None
albert-base-v2-SST	prb_1787	DOUBLE_NEGATIO	POSITIVE → POSITIVE	-6.6	NO	EXPECTED_PRESERVE	None
albert-base-v2-SST	prb_0e0c	INTENSITY	POSITIVE → POSITIVE	+0.2	NO	EXPECTED_PRESERVE	None
albert-base-v2-SST	prb_c3fa	INTENSITY	POSITIVE → POSITIVE	-9.6	NO	EXPECTED_PRESERVE	None
albert-base-v2-SST	prb_eb25	SYNONYM_SUBSTI	POSITIVE → POSITIVE	-4.9	NO	EXPECTED_PRESERVE	None
albert-base-v2-SST	prb_29ea	CONTRAST_NEGAT	POSITIVE → NEGATIVE	+2.8	YES	EXPECTED_FLIP	None
albert-base-v2-SST	prb_f9e4	CONTRAST_POSIT	POSITIVE → POSITIVE	+18.6	NO	EXPECTED_PRESERVE	None
twitter-roberta-ba	prb_52ce	NEGATION_INSER	POSITIVE → NEUTRAL	-33.7	NO	UNEXPECTED_CHANGE	Misweighted
twitter-roberta-ba	prb_1787	DOUBLE_NEGATIO	POSITIVE → POSITIVE	-17.2	NO	EXPECTED_PRESERVE	Misweighted
twitter-roberta-ba	prb_0e0c	INTENSITY	POSITIVE → POSITIVE	+12.6	NO	EXPECTED_PRESERVE	None
twitter-roberta-ba	prb_c3fa	INTENSITY	POSITIVE → POSITIVE	-11.2	NO	EXPECTED_PRESERVE	None
twitter-roberta-ba	prb_eb25	SYNONYM_SUBSTI	POSITIVE → POSITIVE	-1.1	NO	EXPECTED_PRESERVE	None
twitter-roberta-ba	prb_29ea	CONTRAST_NEGAT	POSITIVE → NEGATIVE	-33.2	YES	EXPECTED_FLIP	None
twitter-roberta-ba	prb_f9e4	CONTRAST_POSIT	POSITIVE → POSITIVE	+17.8	NO	EXPECTED_PRESERVE	None
bert-base-uncased-	prb_52ce	NEGATION_INSER	POSITIVE → NEGATIVE	+3.1	YES	EXPECTED_FLIP	None
bert-base-uncased-	prb_1787	DOUBLE_NEGATIO	POSITIVE → NEGATIVE	-2.4	YES	UNEXPECTED_FLIP	Misweighted
bert-base-uncased-	prb_0e0c	INTENSITY	POSITIVE → POSITIVE	+11.8	NO	EXPECTED_PRESERVE	None
bert-base-uncased-	prb_c3fa	INTENSITY	POSITIVE → POSITIVE	+4.8	NO	EXPECTED_PRESERVE	None
bert-base-uncased-	prb_eb25	SYNONYM_SUBSTI	POSITIVE → POSITIVE	+2.6	NO	EXPECTED_PRESERVE	None
bert-base-uncased-	prb_29ea	CONTRAST_NEGAT	POSITIVE → NEGATIVE	+10.5	YES	EXPECTED_FLIP	None
bert-base-uncased-	prb_f9e4	CONTRAST_POSIT	POSITIVE → POSITIVE	+15.3	NO	EXPECTED_PRESERVE	None

Model	Probe ID	Category	Observed Transition	Δ Conf	Flip	Outcome	Diagnosis
twitter-roberta-ba	prb_52ce	NEGATION_INSER	POSITIVE → NEUTRAL	-18.1	NO	UNEXPECTED_CHANGE	**Misweighted**
twitter-roberta-ba	prb_1787	DOUBLE_NEGATIO	POSITIVE → POSITIVE	-6.1	NO	EXPECTED_PRESERVE	**None**
twitter-roberta-ba	prb_0e0c	INTENSITY	POSITIVE → POSITIVE	+4.7	NO	EXPECTED_PRESERVE	**None**
twitter-roberta-ba	prb_c3fa	INTENSITY	POSITIVE → POSITIVE	+0.9	NO	EXPECTED_PRESERVE	**None**
twitter-roberta-ba	prb_eb25	SYNONYM_SUBSTI	POSITIVE → POSITIVE	-4.3	NO	EXPECTED_PRESERVE	**None**
twitter-roberta-ba	prb_29ea	CONTRAST_NEGAT	POSITIVE → NEGATIVE	-19.6	YES	EXPECTED_FLIP	**None**
twitter-roberta-ba	prb_f9e4	CONTRAST_POSIT	POSITIVE → POSITIVE	+19.2	NO	EXPECTED_PRESERVE	**None**